THEORY

Operating systems · five core jobs

The phone is a familiar example of a computer controlled by an operating system. The picture is a memory aid; use the exact technical explanation in exam answers.

WHAT IS AN OS?

An operating system is system software that manages the hardware and resources of a computer and provides an interface and services for applications and users.

1 · USER INTERFACE

Home-screen icons and swipes are how a person gives instructions. A GUI is one form of user interface.

2 · MEMORY + MULTITASKING

The OS allocates RAM to apps and schedules tasks for CPU time. An app switcher is a visible example, not the scheduling mechanism.

3 · PERIPHERALS + DRIVERS

The OS uses drivers to communicate with cameras, speakers, keyboards, printers and other devices, and manages data transfer.

4 · USER ACCOUNTS + ACCESS RIGHTS

The OS can authenticate users and control permissions to settings and data. An Apple ID is not exactly the same as a local OS account.

5 · FILE MANAGEMENT

The OS handles names, folders, saving, copying, moving and retrieving files on storage.

THEORY

Operating systems through an iPhone example

Use the phone examples to connect everyday actions with the OS functions that manage them.



TURN THE PICTURE INTO EXAM LANGUAGE

Select one labelled phone action. Name the operating-system function, describe how it works, then explain why it is useful.

THEORY

What each OS function really does

Keep the analogy, but describe the process rather than just naming a phone feature.

USER INTERFACE

Phone analogy: User taps a camera icon or selects from a menu.

Exam wording: The OS provides an interface for input and output; GUIs, text interfaces and touch gestures are examples.

MEMORY & MULTITASKING

Phone analogy: Phone changes between browser and photos.

Exam wording: The OS allocates RAM and schedules processor time, switching tasks so more than one program makes progress.

PERIPHERALS & DRIVERS

Phone analogy: Phone connects to a printer or processes camera input.

Exam wording: Drivers help the OS communicate with particular devices; the OS manages input/output data transfer.

USER MANAGEMENT

Phone analogy: Device checks a passcode or restricts an app permission.

Exam wording: Account authentication and access rights limit unauthorised use and access to protected resources.

FILE MANAGEMENT

Phone analogy: Phone saves a photo into a folder and moves it.

Exam wording: The OS gives files names, stores and organises them in folders, and enables saving, retrieval and movement.

THEORY

Operating-system functions

This table illustrates several OS functions, including memory, devices and interfaces. File management is explained below.

OS Function	What it does	Key features / examples	Why it matters (exam focus)
User Interface (UI)	Allows the user to interact with the computer	Windows, icons, menus, pointer, touchscreen	Makes the computer usable
Memory Management	Allocates RAM to applications and transfers data between memory and the CPU	Loads programs into memory, frees memory when closed	Prevents crashes and allows programs to run
Multitasking	Allows more than one program to run at the same time	CPU switches quickly between applications	Improves efficiency and user experience
Peripheral Management & Drivers	Controls input/output devices using drivers	Keyboard, mouse, printer, display	Allows hardware to communicate with software
Data Transfer Management	Manages data sent between devices and the processor	Input and output coordination	Prevents data loss or errors
User Management	Creates and manages user accounts	Login accounts, passwords, admin vs standard users	Improves security and controls access
Access Rights	Controls what users can do	Read/write permissions, restricted settings	Protects files and system settings
Security	Protects the system and user data	Passwords, permissions, account control	Prevents unauthorised access

FILE MANAGEMENT

File management organises folders and names, saves and retrieves data, and moves files. The OS manages resources but cannot guarantee that every application will run without errors.

EXAM SCOPE

OCR requires these operating-system functions and their purpose. Detailed implementation of paging and segmentation is not required.

THEORY

Phone-to-OS comparison

Your existing classroom table connects the phone to nine practical OS activities. Below it is the precise distinction to remember.

OS Function	What it does	📱 iPhone Analogy
User Interface (UI)	Allows the user to interact with the computer	The **iPhone home screen**, icons, swipe gestures, and touch
Memory Management	Allocates RAM to applications and transfers data between memory and the CPU	iOS deciding **how much memory each app gets**so apps don't crash
Multitasking	Allows more than one program to run at the same time	Switching between apps using the **app switcher**
Peripheral Management & Drivers	Controls input/output devices using drivers	iOS automatically recognising **AirPods, chargers, printers, keyboards**
Data Transfer Management	Manages data sent between devices and the processor	Touching the screen or taking a photo and iOS **sending that data to the processor**
User Management	Creates and manages user accounts	**Apple IDs** and user profiles on a device
Access Rights	Controls what users can do	**Parental controls** and app permissions (camera, microphone, location)
Security	Protects the system and user data	**Face ID, passcodes, secure boot, encryption**
File Management	Organises and stores data	The **Files app**, folders, saving and moving documents

Exam application – match the OS function: Match each task to its operating-system role: creating a new user profile; allocating RAM to running programs; organising a saved folder; sending data to a printer. Choose from user, memory, file and peripheral management, and explain one match.

WATCH OUT FOR TWO SHORTCUTS

The app switcher shows apps, but scheduling and RAM management are OS processes. An Apple ID manages access to Apple services; the OS also manages local authentication and device permissions. Use the five OCR functions when structuring an answer.

KEY TERMINOLOGY

Key terms for 1.5.1 · 1 of 2

Yellow highlighting indicates key terms; exam definitions explain what each function does.

Operating system

System software that manages hardware and resources, and provides services and an interface for users and applications.

User interface

How a person gives commands to and receives information from a computer (for example, icons, menus, touch or text commands).

Graphical UI (GUI)

A user interface using visual elements such as windows, icons, menus and pointers.

Memory management

The OS allocates RAM to processes and coordinates movement of programs and data through memory.

Multitasking

More than one task makes progress because the OS schedules CPU time and switches between tasks; multiple cores may also run tasks in parallel.

CPU scheduling

Selecting which ready task should run on the processor at a particular moment.

KEY TERMINOLOGY

Key terms for 1.5.1 · 2 of 2

Yellow highlighting indicates key terms; exam definitions explain what each function does.

Peripheral

Hardware used for input, output or storage, such as a keyboard, printer, camera or drive.

Device driver

Software that allows the OS to communicate with and control a particular hardware device.

Data transfer

Movement of data between devices, memory and processor that the operating system helps coordinate.

User account

An operating-system identity with credentials, settings and assigned permissions.

Access rights

Permissions controlling which files, resources and actions a user or program can use.

File management

Creating, naming, saving, organising, moving and deleting files and folders.

WORKED EXAMPLES

Example 1 · Switching between apps

A student records a voice note, opens a browser and returns to the recording.

SCENARIO

An iPhone has a notes app, camera and browser. The student wants to swap quickly between activities without losing their work.

STEP 1 · IDENTIFY TWO OS FUNCTIONS

Memory management allocates RAM to apps. Multitasking lets multiple programs make progress.

STEP 2 · EXPLAIN WHAT THE OS DOES

The OS allocates/reclaims memory and schedules CPU time. When a different app runs, the system switches active work.

STEP 3 · DISTINGUISH THE ANALOGY

The app-switcher screen is visible evidence of multiple apps, but not the mechanism that schedules CPU tasks.

STEP 4 · FORM AN EXAM ANSWER

"The OS allocates RAM and schedules processor time between running tasks, allowing a user to switch between applications."

WORKED EXAMPLES

Example 2 · A school computer

A pupil prints a document but must not edit the teacher-only folder.

SCENARIO

A school laptop connects to a printer. The pupil can print their work but cannot remove files from a protected shared folder.

STEP 1 · PERIPHERAL MANAGEMENT

The OS uses the printer driver to send data and commands to the printer in a format it can use.

STEP 2 · USER MANAGEMENT

The OS recognises/authenticates the pupil account and applies its assigned permissions.

STEP 3 · FILE MANAGEMENT

The pupil names and saves the assignment in a folder. The OS manages its location and retrieval.

STEP 4 · FORM AN EXAM ANSWER

"Drivers allow OS/device communication; access rights prevent the pupil deleting protected files; file management saves and organises the document."

EXAM PRACTICE

Questions · 1 of 4

Answer the questions first; the mark scheme follows later.

EXAM Q1 · 1 MARK

What is an operating system?

EXAM Q2 · 1 MARK

Name one function of an operating system.

EXAM Q3 · 1 MARK

What is a user interface (UI)?

EXAM Q4 · 2 MARKS

State two functions of an operating system.

EXAM PRACTICE

Questions · 2 of 4

Answer the questions first; the mark scheme follows later.

EXAM Q5 · 2 MARKS

State two reasons why an operating system is important.

EXAM Q6 · 3 MARKS

Explain how memory management helps a computer run efficiently.

EXAM Q7 · 3 MARKS

Explain how multitasking is handled by an operating system.

EXAM Q8 · 3 MARKS

Explain the role of device drivers in an operating system.

EXAM PRACTICE

Questions · 3 of 4

Answer the questions first; the mark scheme follows later.

EXAM Q9 · 3 MARKS

Explain how file management helps users.

EXAM Q10 · 4 MARKS

Describe the role of an operating system in managing users.

EXAM Q11 · 4 MARKS

Describe two security features provided by an operating system.

EXAM PRACTICE

Questions · 4 of 4

Answer the questions first; the mark scheme follows later.

EXAM Q12 · 6 MARKS

Explain the main functions of an operating system.

ADDED OCR PRACTICE E1 · 3 MARKS

A student switches between a notes app and the camera. Explain how the OS manages memory and processor time.

ADDED OCR PRACTICE E2 · 3 MARKS

A teacher installs a printer and gives a student account permission to print but not delete shared files. Explain the two OS functions involved.

ANSWERS & MARK SCHEME

Model answers · 1 of 4

These are indicative marking points, not a script that must be memorised word for word.

EXAM Q1 · 1 MARK

What is an operating system?

An operating system is system software that manages a computer's resources (hardware, memory, devices and files) and provides services/user interfaces for programs and people. (1)

EXAM Q2 · 1 MARK

Name one function of an operating system.

Any valid function, e.g. user interface, memory management, multitasking, peripheral/driver management, user management or file management. (1)

EXAM Q3 · 1 MARK

What is a user interface (UI)?

A user interface lets a person interact with or control a computer, providing input and viewing output. (1)

EXAM Q4 · 2 MARKS

State two functions of an operating system.

Any two distinct examples such as memory management (allocates RAM) and peripheral management (controls devices using drivers). (2)

ANSWERS & MARK SCHEME

Model answers · 2 of 4

These are indicative marking points, not a script that must be memorised word for word.

EXAM Q5 · 2 MARKS

State two reasons why an operating system is important.

Two justified reasons: enables users and applications to access hardware/resources (1); manages resources so tasks can run and system operation is coordinated/reliable (1). Avoid claiming crashes are impossible.

EXAM Q6 · 3 MARKS

Explain how memory management helps a computer run efficiently.

The OS allocates available RAM to running programs (1), loads/manages program data in memory (1), and reclaims or reallocates RAM when tasks no longer need it (1). Credit supported points on memory available for multitasking.

EXAM Q7 · 3 MARKS

Explain how multitasking is handled by an operating system.

The OS schedules time on the CPU between runnable tasks (1), saves/restores task state or rapidly switches CPU attention (1), so multiple applications can make progress or appear active simultaneously (1). Multiple cores can also perform tasks simultaneously.

EXAM Q8 · 3 MARKS

Explain the role of device drivers in an operating system.

A device driver provides a software interface between the OS and a particular hardware device (1), translating commands and data as needed (1), and enabling use/control of examples such as printers, keyboards or sound devices (1).

ANSWERS & MARK SCHEME

Model answers · 3 of 4

These are indicative marking points, not a script that must be memorised word for word.

EXAM Q9 · 3 MARKS

Explain how file management helps users.

File management lets users name and save files (1), organise them in folders and move/rename/delete them (1), and retrieve/open the correct stored work efficiently (1).

EXAM Q10 · 4 MARKS

Describe the role of an operating system in managing users.

The OS sets up and manages user accounts (1), authenticates users for access (1), assigns permissions such as read/write or administrator privileges (1), and restricts unauthorised use of resources (1).

EXAM Q11 · 4 MARKS

Describe two security features provided by an operating system.

Examples: authentication using a password/passcode protects access to an account (up to 2); access rights prevent a user reading/changing resources they are not allowed to use (up to 2). Other clearly OS-managed security features with proper explanations may be credited.

ANSWERS & MARK SCHEME

Model answers · 4 of 4

These are indicative marking points, not a script that must be memorised word for word.

EXAM Q12 · 6 MARKS

Explain the main functions of an operating system.

A developed six-mark answer should explain multiple functions: a GUI/CLI lets users interact with the system; memory management allocates RAM and loads data; multitasking schedules processor time; peripheral management uses drivers to control input/output; user management authenticates users and restricts access via permissions; file management names, stores and organises data. Award distinct explained functions rather than a mere list. Do not require paging or segmentation.

ADDED OCR PRACTICE E1 · 3 MARKS

A student switches between a notes app and the camera. Explain how the OS manages memory and processor time.

The OS allocates memory to the active programs (1), schedules processor time between runnable tasks (1), and manages data/state as apps become active or inactive (1). A visible app switcher alone is not the scheduling algorithm.

ADDED OCR PRACTICE E2 · 3 MARKS

A teacher installs a printer and gives a student account permission to print but not delete shared files. Explain the two OS functions involved.

Peripheral management and the printer driver enable the OS to control/send data to the printer (1). User management assigns a student account (1) and access rights restrict deletion while allowing authorised tasks (1).

THEORY

Utility software · extra housekeeping

The operating system manages core computer resources; utilities carry out additional housekeeping and protection tasks, sometimes integrated with the OS.

THE CORE IDEA

Utility software helps maintain, protect or manage a system. OCR J277 focuses on three purposes: encryption, defragmentation and data compression.

ENCRYPTION SOFTWARE

Makes information unreadable to someone without the correct key, protecting confidentiality if a file or device falls into the wrong hands.

DEFRAGMENTATION

Rearranges scattered file blocks on a magnetic HDD to reduce the distance the moving disk head travels and potentially improve access speed.

DATA COMPRESSION

Represents information using fewer bits. It can save storage space and reduce the amount of data that needs transferring.

OS OR UTILITY?

The OS provides core services; utility software handles particular maintenance tasks. Some utilities are built into the OS or run only when requested.

THEORY

Utility software on a mobile phone

Use this comparison to distinguish what the operating system does from the purposes of utility software.

Utility Software	What it does (simple)	Why it is needed	📱 Mobile phone analogy
Encryption software	Scrambles data so only authorised users can read it	Protects personal and sensitive data if a device is lost, stolen or hacked	Locking your phone with **Face ID / fingerprint** so no one else can read your messages
Defragmentation	Reorganises data on a hard disk so files are stored together	Makes files load faster and improves performance on HDDs	Tidying apps and storage so your phone doesn't feel slow
Data compression	Reduces file size by removing unnecessary data	Saves storage space and speeds up file transfer	Your phone **shrinking photos or videos** to save space
Utility software (overall)	Performs extra housekeeping tasks	Keeps the system fast, secure and organised	Background tools that **quietly keep your phone running smoothly**

IMPORTANT: THE ANALOGY IS A MEMORY AID

The Face ID example is authentication (who may access the device), not the encryption algorithm. Tidying phone apps is not the same as HDD defragmentation. Use the following pages for technically accurate descriptions.

THEORY

Three analogies · three real processes

Keep the familiar phone illustration, but distinguish the everyday example from the computing action.

1 · LOCKING A PHONE

Picture/analogy: Face ID or a passcode checks who you are.

Exam explanation: Encryption transforms the underlying information into ciphertext. The device may unlock access to the key after authentication.

2 · TIDYING A DEVICE

Picture/analogy: Rearranging apps on a phone screen makes the interface neat.

Exam explanation: HDD defragmentation relocates separate blocks of file data on a rotating magnetic disk; that is a different process.

3 · SHRINKING A VIDEO

Picture/analogy: The phone can reduce a photo or video file so it takes up less storage.

Exam explanation: Compression uses fewer bits to represent content. Lossless methods preserve data exactly; lossy methods may remove some detail.

THEORY

Encryption protects confidentiality

Use accurate encryption language: the key is the distinction between readable data and data that cannot be

BEFORE · PLAINTEXT

A school stores a document containing student information. In its original form the information can be read by an authorised program.

ENCRYPT · METHOD AND KEY

Encryption software changes the plaintext into ciphertext using an encryption method and key.

IF THE FILE IS INTERCEPTED

An attacker may still obtain or copy the data, but cannot normally understand the protected ciphertext without the appropriate decryption key.

AUTHORISATION AND AUTHENTICATION

A device uses a passcode or biometric check to control access. This can protect access to an encryption key, but the identity check itself is not encryption.

THEORY

Defragmentation · on magnetic HDDs

A simple model: fragments of a single file may be physically separated across the spinning disk.

THE SITUATION

A large file can be stored in separated blocks on an HDD. The moving read/write head may travel to different positions to gather all the data.

BEFORE · FRAGMENTED



A–D are file blocks; X/Y/Z are unrelated blocks. This is a simplified HDD model, not a picture of an SSD.

AFTER · DEFRAGMENTED



A–D are file blocks; X/Y/Z are unrelated blocks. This is a simplified HDD model, not a picture of an SSD.

WHY IT CAN HELP

More contiguous blocks can reduce head movement/seek delays on a rotating HDD. An SSD has no moving read head, so HDD-style defragmentation does not bring the same benefit.

THEORY

Data compression · smaller files

See J277 1.2.5 for lossy/lossless details; here focus on why compression is useful as utility software.

PROBLEM

A large video or backup takes up storage space and may take longer to transfer.

UTILITY ACTION

A data-compression tool encodes the file using fewer bits. Depending on the technique, it might be restored exactly (lossless) or with some detail removed (lossy).

ADVANTAGES

Uses less storage and can transfer faster if fewer bits are sent over a fixed-speed connection. The saving depends on the content, method and settings.

KEY DIFFERENCE

Compression changes file size, not access permission. Encryption changes readability/confidentiality, not necessarily how much storage a file uses.

THEORY

Operating system vs utility software

Study the comparison, then use the technical explanation to prepare an exam answer.

Operating System	Utility Software
Makes the device work	Looks after the device
Manages hardware, memory, users and files	Performs extra maintenance jobs
Always running	Often hidden or runs in the background

WHEN STUDENTS COMPARE THEM

Operating system: provides core services such as memory allocation, device communication, file management and user interface. Utility software: carries out maintenance or protection jobs such as encryption, compression and defragmentation. Utilities may be part of an OS; neither category is defined solely by whether a program runs in the background.

TWO MARKS? GIVE TWO CLEAR DIFFERENCES

Name a relevant OS function (e.g. allocating RAM) and a utility role (e.g. compressing a file). If asked for two differences, develop contrasting roles or examples rather than simply saying “one works and the other cleans”.

KEY TERMINOLOGY

Exam vocabulary · 1 of 2

Yellow highlighting identifies the key terms to use in exam answers.

Utility software

System software that performs additional housekeeping, security or maintenance tasks.

Encryption

Transforming readable plaintext into ciphertext using an encryption method and key.

Plaintext

Data that can be read in its original form.

Ciphertext

Encrypted data that cannot normally be understood without the correct key.

Decryption key

Information needed to convert encrypted data back into readable form with an appropriate method.

Authentication

Checking the identity of a user, such as by a passcode, fingerprint or face recognition.

KEY TERMINOLOGY

Exam vocabulary · 2 of 2

Yellow highlighting identifies the key terms to use in exam answers.

Defragmentation

Rearranging fragmented file blocks on a magnetic hard disk so related blocks are closer together.

Fragmentation

File data stored in separate non-contiguous blocks on a drive.

Hard disk (HDD)

Magnetic storage device whose moving read/write head can incur delays between separated blocks.

Solid-state drive (SSD)

Flash-based storage without a moving read head; routine HDD-style defragmentation is not normally useful.

Data compression

Reducing the number of bits required to represent data, according to a chosen method.

Lossless / lossy

Lossless preserves the exact original when restored; lossy deliberately discards information to reduce size.

WORKED EXAMPLES

Example 1 · The stolen laptop

A school laptop containing pupil data has been lost. Explain how encryption may help.

SCENARIO

The laptop contains a student progress spreadsheet. A thief may possess the laptop but the school has enabled device encryption.

STEP 1 · IDENTIFY THE THREAT

The thief may try to access confidential information stored on the laptop.

STEP 2 · NAME THE UTILITY

Encryption software transforms the protected spreadsheet into ciphertext using a key.

STEP 3 · EXPLAIN THE BENEFIT

Without the correct key, the thief cannot normally interpret the information even if they access a copy of the data.

STEP 4 · AVOID A COMMON ERROR

Face ID/passcodes are authentication controls, not the actual method of scrambling the stored data.

WORKED EXAMPLES

Example 2 · Different utilities for different jobs

Choose the right software for a storage-performance or file-transfer scenario.

SCENARIO A · SLOW MAGNETIC HDD

A school desktop uses a spinning hard disk. Large files have become fragmented across the disk surface.

STEP-BY-STEP HDD ANSWER

1. Choose defragmentation software. 2. It moves separate pieces of a file closer together on a magnetic drive. 3. The read/write head may move less when accessing the file, which can reduce access time. 4. Do not suggest this provides extra disk capacity or is the same as cleaning phone apps.

SCENARIO B · SHARING A LARGE VIDEO

A student needs to send a file over a slow internet connection. The video is larger than necessary.

STEP-BY-STEP FILE TRANSFER ANSWER

1. Choose data compression. 2. It represents the file with fewer bits. 3. A smaller file needs less storage and usually takes less time to send at the same transfer rate. 4. Select lossless if all information must be preserved, or a suitable lossy method where some detail loss is acceptable.

EXAM PRACTICE

Independent questions · 1 of 3

Practice questions and mark values are unchanged. Attempt without looking at the answers on later pages.

EXAM Q1 · 1 MARK

What is utility software?

EXAM Q2 · 1 MARK

Name one type of utility software.

EXAM Q3 · 1 MARK

What is encryption software used for?

EXAM Q4 · 2 MARKS

State two tasks carried out by utility software.

EXAM Q5 · 2 MARKS

State two differences between an operating system and utility software.

EXAM PRACTICE

Independent questions · 2 of 3

Practice questions and mark values are unchanged. Attempt without looking at the answers on later pages.

EXAM Q6 · 3 MARKS

Explain how defragmentation improves performance.

EXAM Q7 · 3 MARKS

Explain why encryption software is important.

EXAM Q8 · 3 MARKS

Explain how data compression benefits users.

EXAM Q9 · 4 MARKS

Describe two examples of utility software.

EXAM PRACTICE

Independent questions · 3 of 3

Practice questions and mark values are unchanged. Attempt without looking at the answers on later pages.

EXAM Q10 · 4 MARKS

Describe the difference between utility software and an operating system.

EXAM Q11 · 6 MARKS

Explain the role of utility software in maintaining a computer system.

ADDITIONAL OCR PRACTICE E1 · 3 MARKS

A student says Face ID is the same as encryption. Explain why this is not quite correct and how the two can work together.

ADDITIONAL OCR PRACTICE E2 · 3 MARKS

A school replaces a magnetic hard disk with a solid-state drive. Explain why running a regular HDD defragmentation utility is no longer an appropriate way to improve speed.

ANSWERS & MARK SCHEME

Indicative model answers · 1 of 3

These are exam-accurate marking points. Equivalent technically correct explanations can receive credit.

EXAM Q1 · 1 MARK

What is utility software?

Utility software performs additional maintenance or housekeeping tasks to help protect or manage a computer. (1)

EXAM Q2 · 1 MARK

Name one type of utility software.

One of: encryption, defragmentation, or data compression. (1)

EXAM Q3 · 1 MARK

What is encryption software used for?

It encrypts readable data into ciphertext using an algorithm/key so someone without the correct key cannot readily understand it. (1)

EXAM Q4 · 2 MARKS

State two tasks carried out by utility software.

Two different tasks, e.g. encrypting information to help keep it confidential (1) and compressing files to reduce the size for storage/transfer (1).

EXAM Q5 · 2 MARKS

State two differences between an operating system and utility software.

OS: manages everyday computer resources and provides core interfaces/services, e.g. managing RAM or hardware (1). Utility software: provides an additional housekeeping or security task, e.g. encryption or compression (1). Other clear differences may receive credit.

ANSWERS & MARK SCHEME

Indicative model answers · 2 of 3

These are exam-accurate marking points. Equivalent technically correct explanations can receive credit.

EXAM Q6 · 3 MARKS

Explain how defragmentation improves performance.

On a magnetic hard disk, fragments of one file may be stored far apart (1). Defragmentation groups those file blocks more closely (1), reducing how far the read/write head moves and potentially reducing access time (1). Do not generalise this benefit to SSDs.

EXAM Q7 · 3 MARKS

Explain why encryption software is important.

Encryption converts readable information into ciphertext (1); if a device is lost, stolen, or data is intercepted (1), someone without the correct decryption key cannot easily read the protected information (1). A phone lock is authentication, not the encryption process itself.

EXAM Q8 · 3 MARKS

Explain how data compression benefits users.

Compression reduces how many bits are needed to represent a file (1), so less storage space is required (1) and smaller files can usually be transferred more quickly at the same network speed (1).

EXAM Q9 · 4 MARKS

Describe two examples of utility software.

Two developed examples: encryption protects confidentiality by turning information into ciphertext that needs a key (up to 2); defragmentation moves separated blocks of a file closer together on an HDD, potentially reducing seek time (up to 2). Compression can also be credited if explained.

ANSWERS & MARK SCHEME

Indicative model answers · 3 of 3

These are exam-accurate marking points. Equivalent technically correct explanations can receive credit.

EXAM Q10 · 4 MARKS

Describe the difference between utility software and an operating system.

The OS provides core resource management and services, such as allocating memory, running tasks or controlling hardware (up to 2). Utility programs perform additional maintenance and security tasks, such as encryption, HDD defragmentation, and compression (up to 2). Some tools are integrated into an OS, so do not define them solely by whether they run in the background.

EXAM Q11 · 6 MARKS

Explain the role of utility software in maintaining a computer system.

A developed answer covers: encryption changes plaintext to ciphertext using a key so untrusted readers cannot understand intercepted or stolen data (up to 2); defragmentation reorganises file blocks on a magnetic hard disk and can reduce read-head movement/access time (up to 2); compression reduces file size and therefore uses less space and transfers more quickly (up to 2). Explain the purpose and usefulness of each rather than simply listing their names.

ADDITIONAL OCR PRACTICE E1 · 3 MARKS

A student says Face ID is the same as encryption. Explain why this is not quite correct and how the two can work together.

Face ID checks the user's identity (authentication) (1). Encryption turns readable data into ciphertext protected by a key (1). A device can use successful authentication to permit access to an encryption key or protected data (1).

ADDITIONAL OCR PRACTICE E2 · 3 MARKS

A school replaces a magnetic hard disk with a solid-state drive. Explain why running a regular HDD defragmentation utility is no longer an appropriate way to improve speed.

An HDD has moving read/write heads whose seek time is affected by separated file blocks (1). An SSD does not have a moving read head (1), so bringing the file blocks together does not provide the same seek-time benefit and creates unnecessary writes (1).